Python Activity-Based Learning
Case Study

53 Food Waste Monitoring System

Selected case study | Prepared using the topics listed in the supplied assignment.

1. Problem Statement

Compare the number of meals prepared, served, and left over across several days. Calculate wastage percentage and identify days that exceed the acceptable wastage limit.

2. Objectives

• Store daily data in lists.

• Calculate leftover meals and wastage percentage.

• Use conditions to flag excessive wastage.

• Summarize results across days.

3. Assumptions

• Data is collected for three days.

• Prepared meals must be greater than zero.

• Meals served cannot exceed meals prepared.

• A wastage percentage above 10% is considered excessive.

4. Algorithm

1. Read prepared and served meals for each day.

2. Validate the values.

3. Calculate leftovers as prepared minus served.

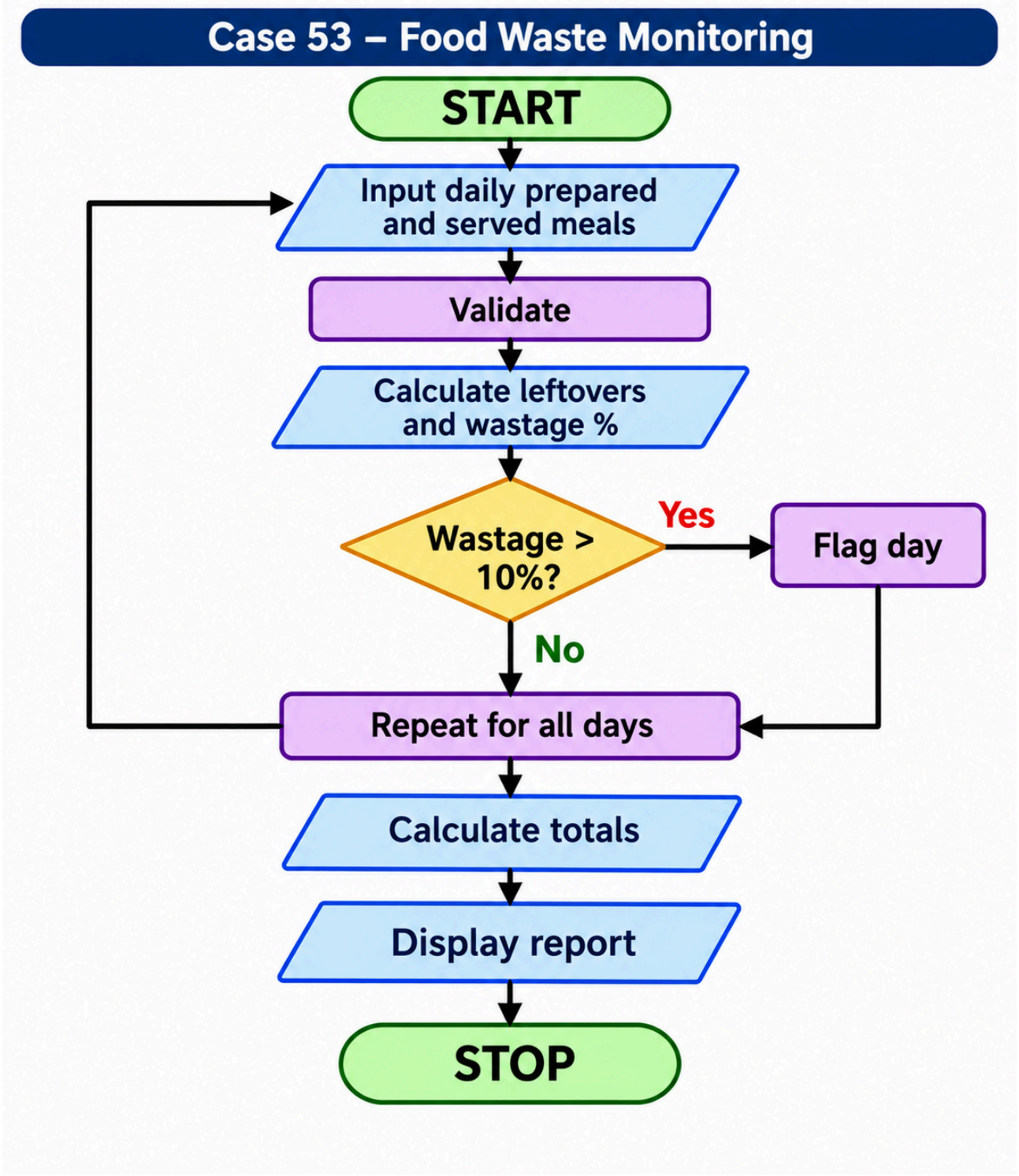
4. Calculate wastage percentage as leftovers divided by prepared times 100.

5. Flag days with wastage above 10%.

6. Calculate total prepared, served, and leftover meals.

7. Display daily and overall summaries.

5. Flowchart



6. Pseudocode

```
BEGIN
 FOR each of 3 days
 INPUT prepared, served
 Validate prepared > 0 and 0 <= served <= prepared
 leftover ← prepared - served
 waste_percent ← leftover / prepared × 100
```

```
 IF waste_percent > 10 THEN flag day
 END IF
 Update overall totals
 END FOR
 PRINT daily and overall summary
END
```

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7. Python Source Code

```
prepared_list = []
served_list = []

for day in range(3):
 prepared = int(input("Meals prepared on day " + str(day + 1) + ": "))
 served = int(input("Meals served: "))

 if prepared <= 0 or served < 0 or served > prepared:
 print("Invalid data. Enter this day again.")
 prepared = int(input("Meals prepared: "))
 served = int(input("Meals served: "))

 prepared_list.append(prepared)
 served_list.append(served)

total_prepared = 0
total_served = 0
total_leftover = 0

print("\n--- FOOD WASTE REPORT ---")
for i in range(3):
 leftover = prepared_list[i] - served_list[i]
 waste_percent = leftover / prepared_list[i] * 100

 total_prepared += prepared_list[i]
 total_served += served_list[i]
 total_leftover += leftover

 print("Day", i + 1, "Leftover:", leftover,
 "Wastage:", round(waste_percent, 2), "%")
 if waste_percent > 10:
 print("Warning: excessive food waste")

overall_percent = total_leftover / total_prepared * 100
print("\nTotal prepared:", total_prepared)
print("Total served:", total_served)
print("Total leftover:", total_leftover)
print("Overall wastage:", round(overall_percent, 2), "%")
```

8. Test Cases

Input / Scenario	Expected result
Prepared 100, served 95	Leftover 5; wastage 5%; no warning
Prepared 100, served 80	Leftover 20; wastage 20%; warning
Prepared 50, served 50	Leftover 0; wastage 0%

9. Conclusion

The program applies lists, loops, arithmetic, and conditions to monitor food wastage and identify days requiring

attention. Note: Run the program in Python 3. The sample test cases are illustrative; enter the values when prompted.